



Abdulmalik Abdulmawla

SPATIAL-DATA TOOLS: ARCHITECTURE, URBAN AND RURAL SPACE, MOBILITY, CITY ECONOMY

PROFILE

I come from research and have been building tools for many years that help solve spatial problems: for the design, generation and analysis of spatial data. I believe that any complex challenge, and any handling of data, requires intelligently breaking the problem down to its core parameters and simplifying each element until a humanly manageable scale is reached; this is how nature works: complex systems are not made of complex parts, but of very simple ones, intelligently put together. Technology is one of my strengths; my main one is this computational thinking about spatial data problems, together with the capacity to work my way into the field the data serves, from simple to complex objectives. In doing so I work on the systems behind the tools, from agent-based simulation to AI-assisted, multi-agent coding.

EXPERIENCE

Dec 2023 – Sep 2026

Research Associate, Chair of Transport System Planning

Bauhaus-Universität Weimar · Bauhaus.Mobility / European Digital Innovation Hub Mid Germany (formerly EDIH Thuringia)

Interactive geodata platforms, among them the browser-based analysis platform for Germany's official accident data (sole developer; the official M-Uko 2012 method as an interactive tool); data-science consulting and training for public administrations in open geodata; supervision of student projects.

May 2016 – Mar 2023

Research Associate, Computer Science in Architecture (InfAR, now InfAU)

Bauhaus-Universität Weimar

Teaching computational methods for urban planning and analysis; development of design and research tools (DeCodingSpaces Toolbox, mineR); supervision of master's theses; conference papers (Space Syntax Symposium, eCAADe).

Dec 2017 – Jan 2019

Project Coordinator, DAAD exchange project "Discovering Cities"

Bauhaus-Universität Weimar and German-Jordanian University, Amman · held concurrently with the InfAR post

Project proposal, project finances, teaching seminar in Weimar, workshop in Amman.

May 2016 – 2020

Contributing Author, DeCodingSpaces Toolbox

DecodingSpaces GbR, Weimar

C# development, testing, documentation and consulting; interactive masterplan-analysis tool for Blankenburger Süden (TSPA, Berlin, 2019).

EDUCATION

Nov 2018 – present

PhD candidate, in progress · Computer Science in Architecture and Urbanism (InfAU), Bauhaus-Universität Weimar

Working title: "Spatial dependency of urban retail: detailed methods for positioning street-level activities" · supervisor: Prof. Dr. Reinhard König

2013

Master of Arts in Architecture · Hochschule Anhalt, Dessau-Roßlau, Germany

Thesis: "Heterogeneous-Atelier Industry" (with Ali Farhan): Şişhane's atelier networks, mapped and simulated as one bottom-up factory

2007

BSc Architecture and Engineering · Mansoura University, Egypt

CONTACT

LOCATION

Weimar, Germany

EMAIL

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PORTFOLIO

abdulmalikabdulmawla.github.io

GITHUB

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LANGUAGES

Arabic · native

English · fluent

German · very good

SKILLS

Spatial data and analysis

Official German geodata (ALKIS, cadastre, LoD2/CityGML, WFS/OGC), OpenStreetMap, spatial SQL (DuckDB, GeoParquet), QGIS, network analysis, space syntax, statistics (R)

Tool and application development

Python, TypeScript/JavaScript (React, deck.gl), C# (Grasshopper), simulation and agent-based models, Git, continuous integration, AI-assisted development (Claude Code)

Teaching and consulting

Seminars, workshops, training public administrations in working with open geodata

Design and visualisation

Rhino, AutoCAD, Revit, Adobe Photoshop, Illustrator, InDesign

TEACHING

- 2023 – 2026** · Digital Traffic
Simulation Lab · Open Mobility
Data · Rent-a-Data-Scientist ·
EDIH+BM.VSP.BUW
- 2016 – 2023** · Algorithmic
Architecture · Parametric Urban
Design and Analysis · SYNCITY I
and II · Circular Urbanism ·
InfAU.BUW
- 2018** · Discovering Cities teaching
seminar (DAAD) · InfAU.BUW
- 2014 – 2016** · Design 5 Studio ·
Web Application in Architecture
· AR.AASTMT
- Supervision of master's theses
and projects · 2016 – 2023
InfAU.BUW · 2023 – 2026
DTRL.VSP.BUW

WORKSHOPS (ORGANISER)

- 2018** · Discovering Cities
Workshop · GJU, Amman
- 2018** · Urban Analysis, Synthesis
and Exploration with
Grasshopper · eCAADe, Łódź
- 2016** · Rural-to-Urban
Transformation Workshop · ECL,
Addis Ababa
- 2015** · Simulating Agent-Based
Systems using Processing ·
Cairo

RECOGNITION

- 2026** · Ten-year service
recognition at the Bauhaus-
Universität Weimar, presented
by the University President

COMPETITIONS

- 2014** · Lusail Underpasses Public
Art Competition, phases 1 and
2, Doha
- 2007** · 2nd place, 4th National
Architecture Students
Competition, Cairo University
- 2006** · 1st place, 3rd National
Architecture Students
Competition, Cairo University

BUW Bauhaus-Universität Weimar · VSP
Verkehrssystemplanung · InfAU
Informatik in der Architektur und
Urbanistik · EDIH+BM European Digital
Innovation Hub Mid Germany,
Bauhaus.Mobility · DTRL Digital Transport
Research Lab · AASTMT Arab Academy
for Science, Technology and Maritime
Transport, Cairo · AR Architecture
Department · GJU German-Jordanian
University · ECL Emerging City Lab

EXPERIENCE (CONTINUED)

**Oct 2014 – Apr
2016**

Research Associate, Teaching and Computational Design

Arab Academy for Science, Technology and Maritime Transport
(AASTMT), Cairo

Teaching computational and parametric design (Grasshopper,
Processing, Arduino) in the RIBA Part I programme.

**Jan 2012 – Apr
2016**

Co-founder and Architect, Gozour Studio

Mansoura, Egypt

Residential, education and competition projects, among them the Lusail
Underpass Competition, Qatar.

**May 2009 – Sep
2011**

Junior Architect, EAI Group, Abu Dhabi

Elite Design and Engineering Consultancy, semi-government: client
representative for outsourced projects, client consultant for in-house
projects

**Jul 2008 – Apr
2009**

Freelance Architect, Egypt

3D modelling and visualisation for residential construction projects

**Jul 2007 – Jun
2008**

Junior Architect, Invert Studios, Cairo

Architectural design and planning at the office of Kamel Louqman and
Amir Wahid

Jun – Aug 2005

Intern, Al Turath Engineering Consultancy, Abu Dhabi

CAD design, layout and plotting of complete project sets

SELECTED PUBLICATIONS

- 2022** **Abdulmawla, A.**, Bielik, M., Schneider, S., König, R. Mapping active frontages: a method for linking street-level activities to their surrounding urban forms. A case study of the city of Weimar, Germany. *13th International Space Syntax Symposium*, Bergen.
- 2022** **Abdulmawla, A.**, Schneider, S., König, R., Bielik, M., Fuchkina, E. Parametric urban data structuring and spatial query: advanced data mapping and selection methods for parametric modelling environments. *40th eCAADe*.
- 2019** Bielik, M., König, R., Fuchkina, E., Schneider, S., **Abdulmawla, A.** Evolving configurational properties: simulating multiplier effects between land use and movement patterns. *12th International Space Syntax Symposium*, Beijing.
- 2018** **Abdulmawla, A.**, Schneider, S., Bielik, M., König, R. Integrated data analysis for parametric design environments: mineR, a Grasshopper plugin based on R. *36th eCAADe*.
- 2017** König, R., Bielik, M., Knecht, K., **Abdulmawla, A.**, Fuchkina, E. New methods for urban analysis and simulation with Grasshopper: using the DeCodingSpaces-Toolbox. *CAAD Futures 2017*, Istanbul.
- 2017** Schneider, S., **Abdulmawla, A.**, Donath, D. The effect of the street network on movement patterns and land use in small cities: a comparative study of three 10K cities in Thuringia, Germany. *11th International Space Syntax Symposium*, Lisbon.
- 2017** Dennemark, M., Schneider, S., König, R., **Abdulmawla, A.**, Donath, D. Towards a modular design strategy for urban masterplanning: experiences from a parametric urban design studio on emerging cities in Ethiopia. *eCAADe 2017*, Rome.
- 2016** **Abdulmawla, A.**, Farhan, A. Heterogeneous assemblages of atelier industries. *SimAUD 2016*, London (poster).

Full list: researchgate.net/profile/Abdulmalik-Abdulmawla